

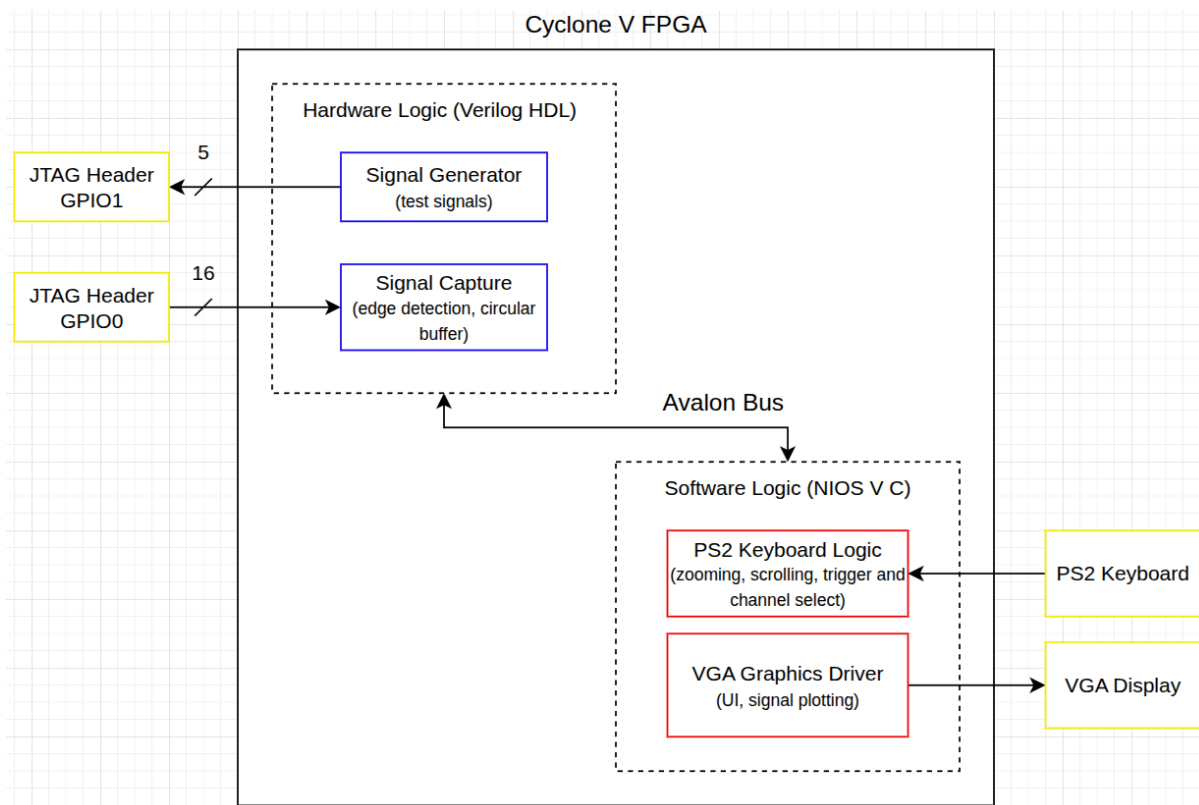
ECE243 Final Project: Logic Analyzer

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Project Description:

Logic Analyzer that samples digital signals on 16 channels from the GPIO expansion header. It polls the input pins to capture signal data and stores it in a circular buffer. The signals are plotted on the VGA and the display is controlled with a PS2 keyboard. Features include adjustable time division and vertical scale, time measurement cursor, and selectable trigger channel. The signal capture logic was implemented on the FPGA hardware as a memory mapped IO (Avalon MM slave) and the display logic runs on the Nios V processor.

Block Diagram:



UI Overview – Logic Analyzer:

The logic analyzer supports visualisation and interaction using the VGA and a PS/2 Keyboard. The system supports visualization of 16 channels total, separated as 8 channels on 2 pages which can be navigated using the Tab key.

Table 1: UI features summarized

Top status bar:	• Top status bar displays: ○ Time per division (ns/div) ○ Sampling frequency (MHz) ○ Number of visible samples ○ Time offset from the trigger point (ns)
Input Controls:	• Supported keys: ○ S, T, E, C, Esc, Tab (basic controls) ○ 0–7 (channel selection) ○ +, – (zoom control) ○ Arrow keys (navigation)
Channel Selection:	• Active channel displayed on HEX0 & underlined on UI: ○ Range: [0–7] • No selection state: ○ Esc or navigate off-screen ○ HEX display cleared • Navigation: ○ ↑/↓ to move between channels ○ Keys 0-7 for direct channel selection • Channel control: ○ Space or E for toggle enable/disable of channels on screen
Control Functions:	• Channel control (must enable signal to trigger and show) ○ Space or E: enable or disable the selected channel • Trigger ○ T: set trigger channel based on current selection • Start data capture and visualize on screen: ○ S: start signal acquisition and waveform display • Clear ○ C: clear all waveforms and reset state
Zoom & Time Scaling:	• Zoom controlled with + and - • Zoom levels (by visible samples): {64, 96, 256, 512, 1024} • Time/div dynamically computed from: ○ Sampling frequency ○ Visible sample window
Time Axis:	• Horizontal time scale rendered across screen • Range: [0, 39,000] ns • Computed from buffer size and sampling rate

Logic Analyzer Parallel Port:

Address	31 ... 16	15 ... 3	2	1	0	Registers
0x80400000	Unused				RUN	Control
0x80400004	Unused		TRIG	FULL	RUN	Status
0x80400008	Unused	TRIGGER_CHANNEL [15:0]				Trigger Config
0x8040000C	Unused	BUFFER_DATA [15:0]				Buffer Window
0x80400010	Unused	TRIGGER_PTR [15:0]				Trigger Pointer
0x80400014	POST_COUNT [15:0]	PRE_COUNT [15:0]				Samples

Control: Write a 1 to the RUN bit in the idle state to start sampling. Write a 0 to the RUN bit after the FULL bit goes high to reset to the idle state.

Status: Read-only. RUN: high when sampling, low when in idle state. FULL: high when the internal buffer is full. TRIG: high after the rising edge trigger condition is detected.

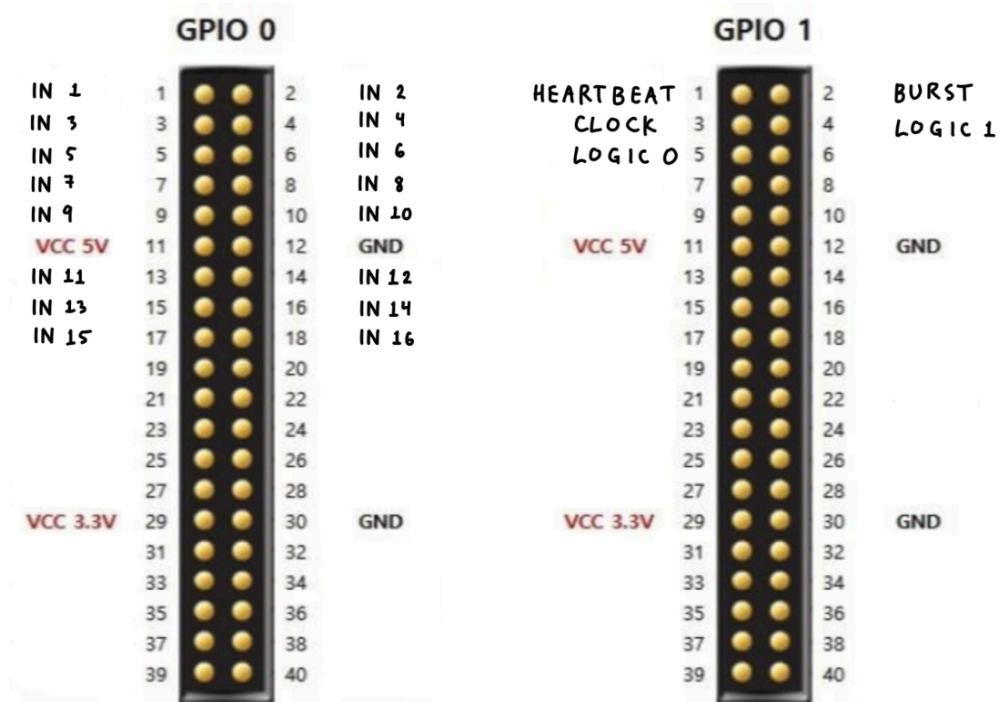
Trigger Config: Write a 16 bit unsigned integer to set the channel to trigger on.

Buffer Window: The bottom 16 bits return the value of the buffer at the internal read_pointer index. Reading from this register auto-increments the internal read_pointer. Writing any value to this register resets the read_pointer back to 0.

Trigger Pointer: Read-only. Returns a 16 bit unsigned integer corresponding to the buffer index when the trigger condition was detected.

Samples: Read-only. The top 16 bits encode the number of post-trigger samples collected. The bottom 16 bits encode the number of pre-trigger samples collected.

JTAG Header Pin Assignments:



Attribution Table:

Section	David Ceuca	Prithika Paraneetharan
PS/2 Keyboard (C)		✓
VGA and graphics (C)		✓
Signal Capture and Signal Generator modules (SystemVerilog)	✓	
Logic Analyzer HAL (C)	✓	
Integration + Debugging	✓	✓